



# Mo<sub>2</sub>CT<sub>x</sub> MXene — Multilayer Powder

Research-grade two-dimensional transition-metal carbide powder

Mo<sub>2</sub>CT<sub>x</sub> is a two-dimensional transition-metal carbide MXene prepared from a Mo<sub>2</sub>Ga<sub>2</sub>C precursor. This multilayer powder provides a layered, micron-scale research material for electrochemical, catalytic and interfacial studies. Surface terminations include –O, –OH, and –F, with distributions dependent on synthesis and lot.

## KEY FEATURES

- Layered Mo<sub>2</sub>CT<sub>x</sub> morphology derived from Mo<sub>2</sub>Ga<sub>2</sub>C
- Micron-scale nominal flake / particle size
- Typical conductivity of 10–100 S/cm
- Mixed –O/–OH/–F surface terminations

## POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal–sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

## STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

## QUALITY / DOCUMENTATION

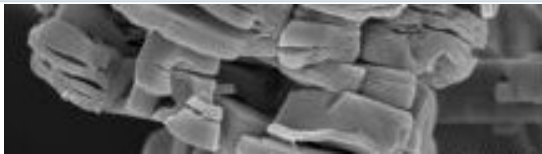
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

## TECHNICAL SPECIFICATIONS

| PROPERTY                      | TYPICAL / REPRESENTATIVE VALUE        |
|-------------------------------|---------------------------------------|
| Material                      | Mo <sub>2</sub> CT <sub>x</sub> MXene |
| Precursor                     | Mo <sub>2</sub> Ga <sub>2</sub> C     |
| Product form                  | Multilayer powder                     |
| Preparation                   | HF etching                            |
| Layer structure / count       | Multilayer                            |
| Nominal flake / particle size | 2–20 μm                               |
| Typical conductivity          | 10–100 S/cm                           |
| Surface terminations          | –O, –OH, and –F                       |
| Appearance                    | Dark powder (descriptive only)        |
| Recommended storage           | Dry; vacuum or inert atmosphere       |

*Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.*

## REPRESENTATIVE CHARACTERIZATION

|                                                                                     |                                                                                                      |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|  | Multilayer Mo <sub>2</sub> CT <sub>x</sub> — SEM. Representative characterization; not lot-specific. |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|

**Product positioning:** Supplied and technically supported by WiniGen Materials for research and development use.

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MO2CTX-ML-TDS-001

Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.